WATER FILTRATION
AMR ENGINEERING WORKS

ABOUT **AMR ENGINEERING WORKS**



- **AMR ENGINEERING WORKS IS ESTABLISHED IN THE YEAR 2009 AS A WATER FILTRATION TECHNOLOGY & ENGINEERING SOLUTION. THE OFFICE OF THE COMPANY IS SITUATED IN THE KHOPOLI.**
- **FOR MORE THAN 16 YEARS OF FIELD EXPERIENCE & DOMAIN EXPERTISE INTO WATER & WASTE WATER FILTRATION TECHNOLOGY, ENGINEERING SOLUTION, DEVELOPING MANY MORE SUCCESSFUL APPLICATIONS FOR INDUSTRY, MUNICIPAL & AGRICULTURAL MARKET. DELIVERING HIGH PERFORMANCE SOLUTIONS FOR EXTREME CONDITIONS WITH EXCEPTIONAL RESULT ALONG WITH 24X7 SERVICE SUPPORT.**
- **THE COMPANY HAS A TEAM OF QUALIFIED ENGINEERS AND DRAFTSMEN TO PROVIDE DETAILED ENGINEERING SERVICES FOR THE WATER FILTRATION SYSTEM.**









Municipal



Industry



Irrigation

To become the global market leader by offering world class product on Water Treatment and Filtration Technology starting from automatic, semiautomatic, high end glass media filtration, screen, disc, microfiber technologies down to <1 micron under one roof…

OUR COMMITMENT PROTECT ENVIROMENT

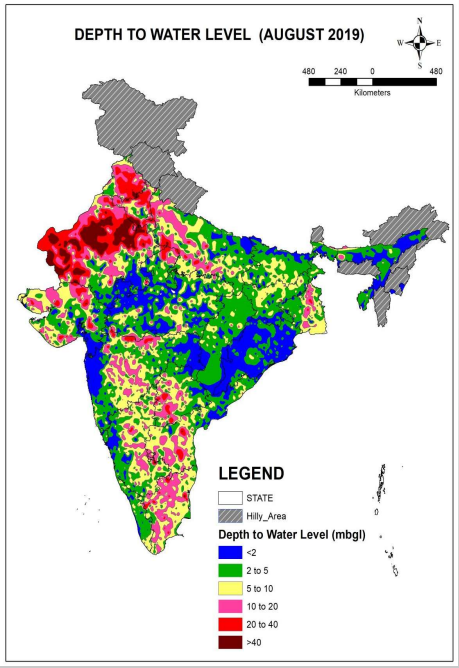


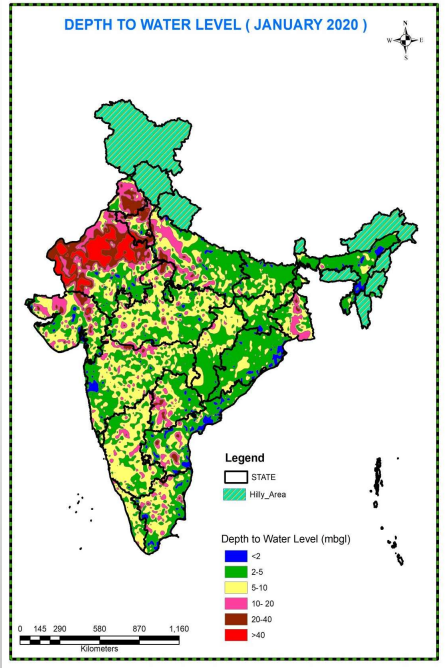


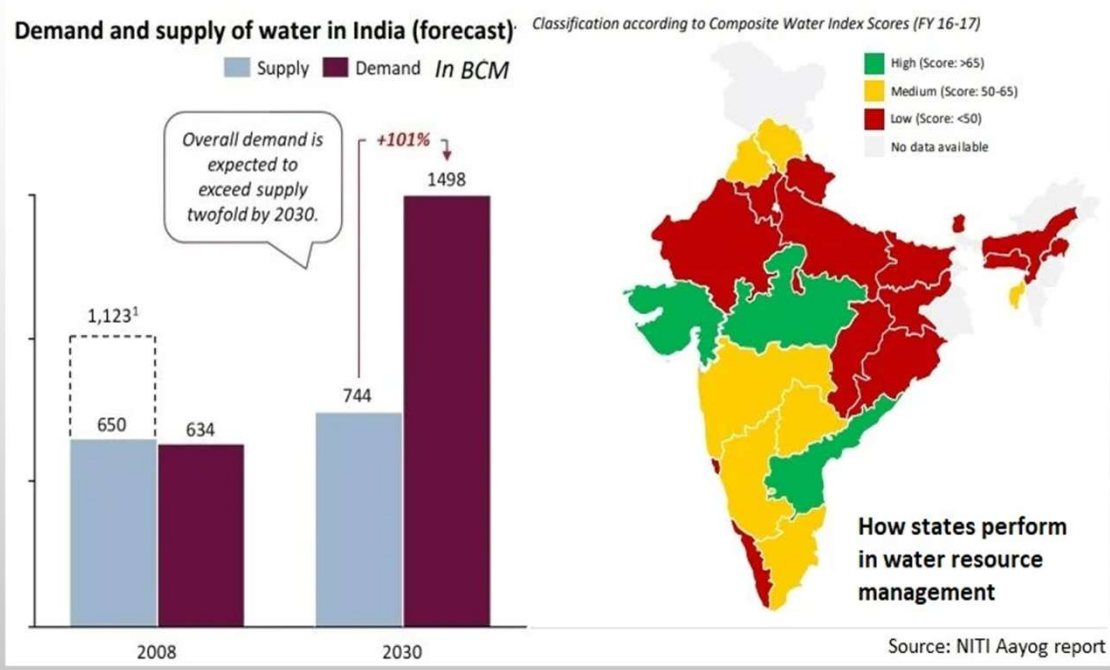



Small
Footprint
Green
Technology
Low
Maintenance
Saving
Energy
Return on
Investment
Commitment
Saving
Water

WATER LEVEL, SUPPLY & DEMAND IN INDIA

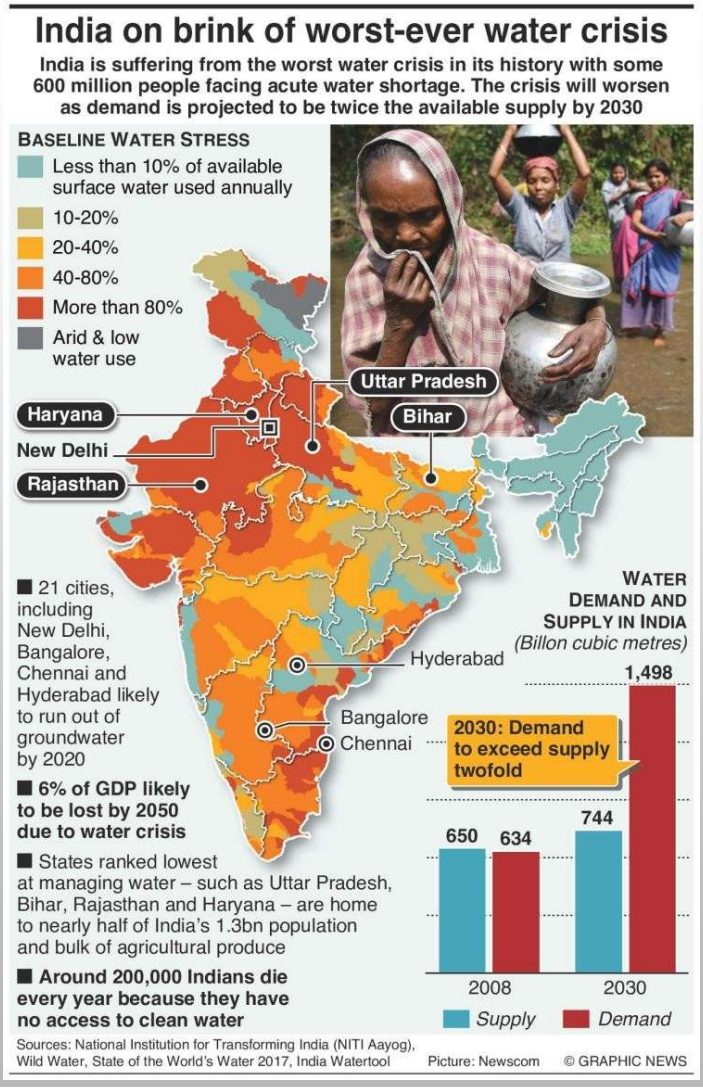


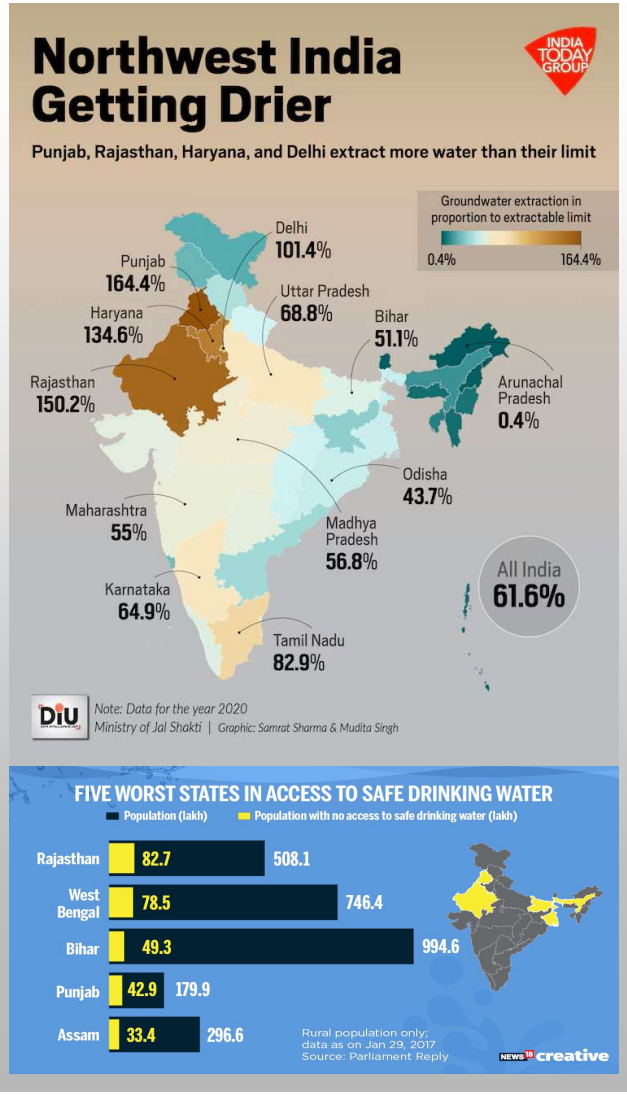


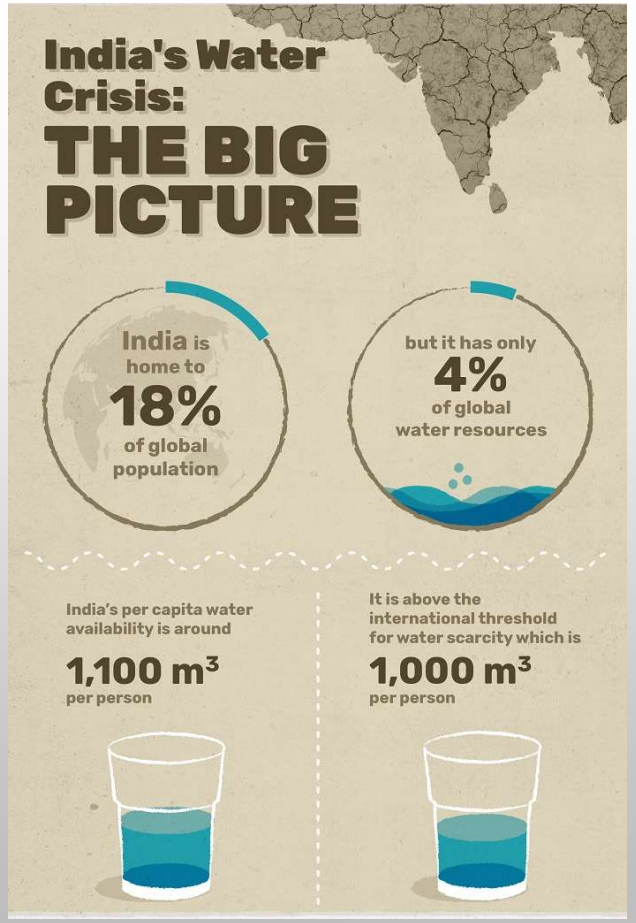


We are committed to protect environment through filtration technology for safe water and also save the water with recycle & reuse process.

WATER LEVEL, SUPPLY & DEMAND IN INDIA







INTRODUCTION
PRIMARY FILTRATION

WHAT IS PRIMARY FILTRATION

Primary water filtration is the initial stage of water treatment that removes large, suspended solids like debris, sediment, and particles from water. It serves as the first step in making water safer and clearer for subsequent treatment or use by preventing large contaminants from clogging or damaging more sensitive filtration stages.

Purpose and function

Removal of large debris: The primary goal is to remove coarse materials like sticks, leaves, and large particles that would interfere with later treatment steps.

Physical barrier: It uses a physical barrier to trap suspended solids that would otherwise make the water cloudy and could cause problems in later stages.

Improved efficiency: By removing large solids upfront, primary filtration protects the effectiveness and extends the life of more complex and expensive filters that come later in the process.

SCREEN FILTERS



The FLOSAM Screen filters work well with inorganic or some organic matter such as large debris, leaves, algae, shrimps, silt and sand. They generally consist of a stainless-steel mesh that traps particles. Screen filters are generally cheaper to purchase than disc or media filters. Consists of heavy-duty automatic filters with an electric self-cleaning mechanism.

Filtration Process:

Raw water enters the inside of the screen cylinder (1) from the filter's inlet (2) and flows through the screen to the filter's outlet (3). The dirt particles are trapped on the inner screen surface and formulate a "filtration cake" that causes a differential pressure across the screen.

Self cleaning Process:

During the self-cleaning process, filtered process water continues to flow. The Exhaust valve (4) opens, and the Drive unit (5) rotates two stainless steel (6) which sweep the inner surface of the cylindrical screen. The particles trapped on the screen are dislodged by the brushes and flushed out the exhaust valve.

SCREEN FILTERS

Control System:

The control system is comprised of a Pressure Differential Switch (7) that senses the pressure Differential across the screen and sends a signal to the control board when it reaches a pre-defined value (usually 0.5 bar / 7 psi). The control board initiates the self-cleaning process.

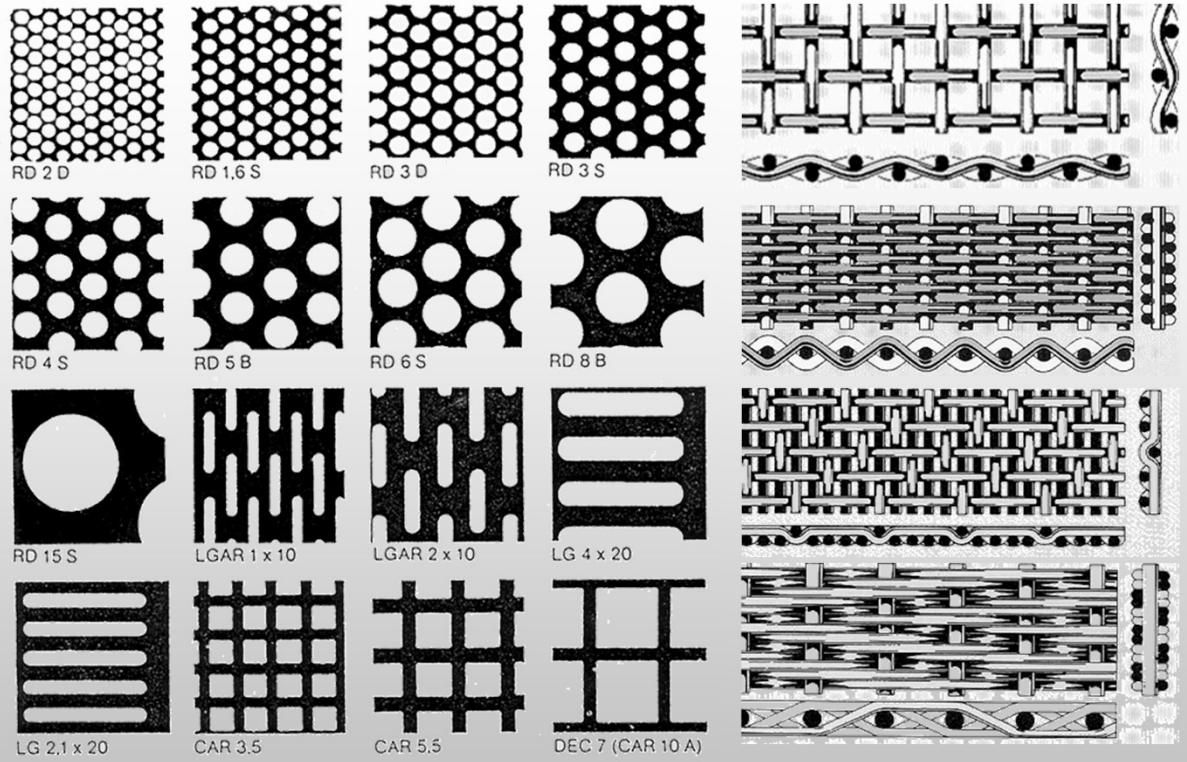
The filter begins a self-cleaning cycle under any one of the following conditions:

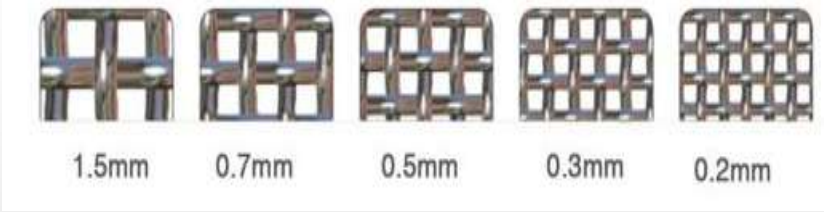
1. Receiving a signal from the Pressure Differential Switch
2. Time interval parameter set at the control board
3. Manual start

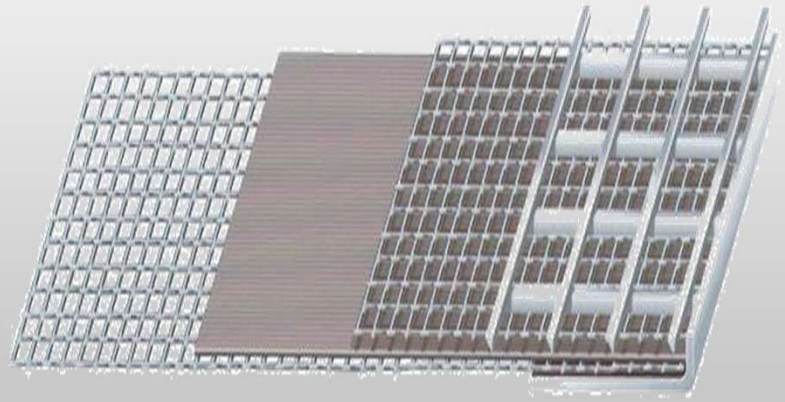
Advantages of Primary Filter:

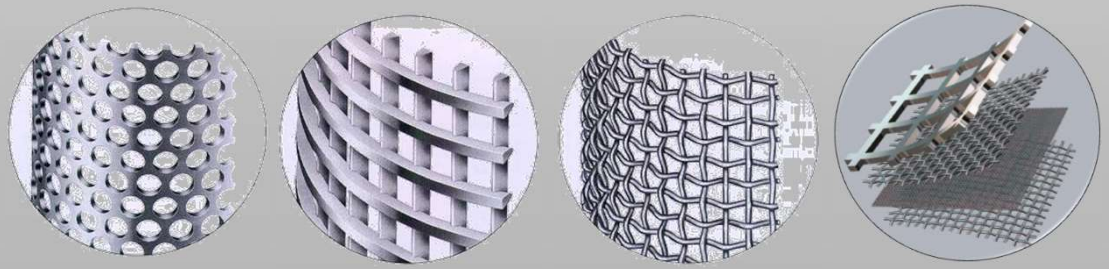
1. Minimum head loss
2. Online filtration i.e. filtration does not stop irrespective of the backwash
3. Less waste of water (< 1% of total flow)
4. Reduced footprint area
5. Energy saving as there is no pump required for backwash
6. Largest filtration area for each filter
7. 100% Uniform cleaning
8. Performance guarantee on the outlet water quality

SCREEN TYPES





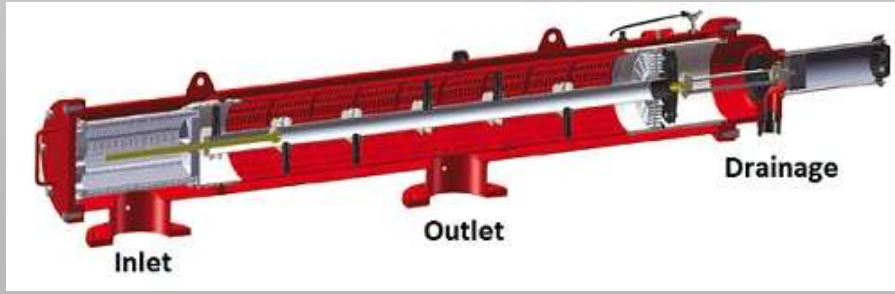


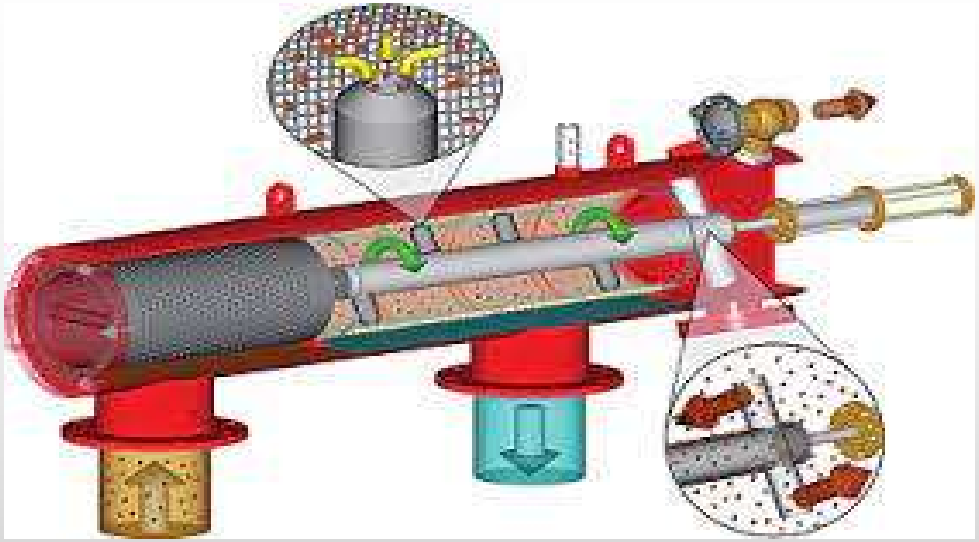


HYDRAULIC FILTERS









The HDLSAM Hydraulic filters are mid-range automatic filters, ideal for remote installation sites, with a water-driven self-cleaning mechanism that doesn't require external power source to operate. With their various screen areas, the Hydraulic filters support flow-rates of up to 400 m^3/h, with filtration degrees of 500 down to 80 micron and inlet/outlet diameters from 2"-10".

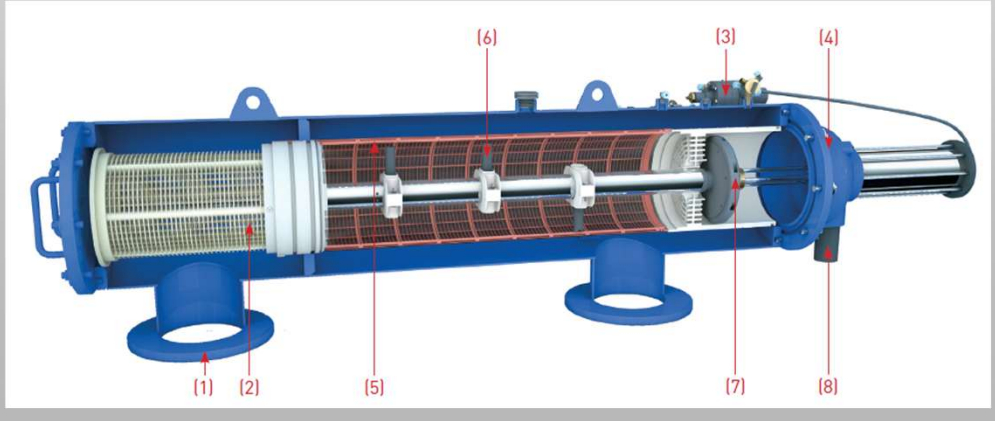
HYDRAULIC FILTERS

The Filtering Process:

The filtration process begins when raw water flows through the filter inlet (1) into the coarse screen (2). Here, the water is pre-filtered in order to protect the cleaning mechanism from large debris. The water then passes through the inner surface of the fine screen; dirt particles are trapped and accumulate inside the filter while clean water flows through the filter outlet. The gradual dirt buildup on the inner screen surface causes a filter cake to develop, with a corresponding increase in the pressure differential across the screen.

The Self-Cleaning Process

When the pressure differential across the screen reaches a pre-set level of 0.5 bar (7 psi), the Hydraulic filter's rinse controller starts the cleaning process by opening the internal flush valve (4). This results in the release of a back-flush stream flowing through the nozzles (5) out of the hollow Suction Scanner Shaft (6) and the Turbine (7) to the Drainage pipe (8).



HYDRO-CYCLONE FILTERS



HCFSAM Hydro cyclones use centrifugal force to "spin" sand out of water. Often water from bores contains high levels of sand. The hydro cyclone creates a spinning action causing the heavier sand particles to separate from the water and become trapped in a storage tank at the base of the unit. Hydro cyclones are generally used in conjunction with another type of filter.

Hydrocyclones are widely used upstream of a filter to first remove the bulk of the contaminant, where the filter does the final cleaning. This type of design reduces the time required to flush and clean the filter. Hydrocyclones are very effective tools to separate solids from liquids.

If to correct the common mistake in the market;
A Hydrocyclone Separator is not a true filter, since there is no screen element inside to filter out the particles...

Some of the Application Areas:

Cooling Towers
Hydropower Plants
Process Water Treatment
Iron & Steel Industry
All Kinds of Industries where density dependant separation is possible

HYDRO-CYCLONE FILTERS



Cyclone Separators focus on the separation of settleable particles – especially having a specific gravity more than 2 gr/cm3

Bigger the specific gravity and particle sizes; better the separation efficiency.

There is not a limit regarding the dirt load at the Inlet.

Settled particles are purged out against time.

Minimum 1 Bar pressure is required for the efficient operation of Cyclone Separators.

if the flow rate gets to a level lower than the operational range of the installed separator then it would operate with a lower efficiency (approx. 20%).

If the flow rates higher than the operational range then separation efficiency would drop dramatically – approx. 50%.

Auto purge system operates according to; purging interval and purging duration. Both can be adjusted based on the actual site conditions.

HYDRO-CYCLONE FILTERS

Separation is ensured by means of the centrifugal force. Separation efficiency is directly related with;
* Specific gravity and,
* Size distribution of the particles

Advantages of Hydrocyclones:

Separation without any interruption of flow

Very low water loss - ~%1-2 of total flow

Very low Pressure Drop - ~0.3 Bar

No need for any consumable material

No need for maintenance

Filtration down to 50 Micron depending on the specific gravity of the particles

Capacities from 1 to 4000 m3/h

Very easy installation & operation

INTRODUCTION
SECONDARY FILTRATION

WHAT IS SECONDARY FILTRATION

Secondary water filtration refers to the stage of treatment that follows primary, physical, or coarse filtration, focusing on removing finer suspended solids, dissolved organic matter, or specific contaminants. It is widely used in wastewater management, industrial processes, and advanced domestic filtration systems to improve water quality and protect downstream equipment.

Purpose and function

Further Purification: It removes smaller impurities, dissolved materials, and fine particles (down to sub-micron levels) that pass through primary filtration, ensuring clearer water.

Protection of Downstream Equipment: In industrial applications, secondary filters prevent fouling, clogging, and scaling in sensitive machinery, such as heat exchangers, reverse osmosis membranes, and boilers, thus extending equipment lifespan.

Enhancing Water Safety: It acts as a crucial barrier against pathogenic organisms and pathogens that can cause disease, improving overall water safety.

MIXSAM

STATIC MIXER



MIXSAM static mixer is an inline device that continuously blends two or more fluid streams (liquids or gases) without moving parts. It uses fixed internal elements to divide and recombine flow, creating homogeneous mixtures with low energy consumption and low maintenance. They are essential in industries like chemical processing, water treatment, and adhesive dispensing.

Mechanism: In laminar flow, fluids are split into layers while turbulent flow uses vortex generation for radial mixing.

Design: Consists of a housing (pipe) containing stationary elements (helical or plate-type).

Construction Materials: Available in materials such as 316 Stainless Steel, MS to suit chemical compatibility.

No Moving Parts: Lowers maintenance requirements and eliminates the need for external motors.

AMR

LAMELLA PLATE SETTLERS



ECOSAM Lamella plate settler are high-rate gravity sedimentation devices designed to remove suspended solids from liquids. They are used extensively in water and wastewater treatment to provide a large settling surface area within a compact footprint, often reducing the necessary tank size by 80% to 90% compared to conventional clarifiers.

High-Efficiency Separation: Uses inclined plates (50˚ to 60˚) to promote laminar flow.

Compact Design: Ideal for applications with limited space; often modular for easy expansion.

Mechanism: Particles settle onto the plates and slide down to a hopper, often aided by flocculant addition for higher turbidity water.

High Reliability: Minimal moving parts allow for lower maintenance compared to other clarifying technologies

GLASS MEDIA FILTERS



AQUALIFE Media filters are suitable for both inorganic and organic contaminants. They are tanks filled with activated glass media, sand or gravel media. Water passes through the bed of media and particles are trapped. Media filters offer a much bigger surface area and as such can usually provide more effective filtration. Cleaning is achieved by reversing the flow of water through the tank to lift and separate the glass or sand. Trapped contaminants are purged to waste as the water flows through.

In AQUALIFE filters we are used advance activated glass media for best performance and life long solution. The active glass media is made from crushed green glass & hydraulically optimized in terms of gain shape & size to enhance performance.

The surface area of glass media is 10,00,000 m2/m3 & it is 300 times of the sand surface area. The zeta potential of glass media is > - 70mV. & for sand -10mV to -30mV.

GLASS MEDIA FILTERS



Under typical environmental and water treatment conditions (pH 5–9), the MEDIA surfaces are negatively charged, which creates a barrier to removing negatively charged particles, bacteria, and organic matter.

The activated glass media is eco-friendly & sustainable solution with high performance like 80% removal of all particles in the water down to 4.5 microns.

The activated glass media removes organic & inorganic impurities. Some BOD & COD reduction also done.

The best part of this glass media is no formation of Bio-film, no mud ball formation so no need to replace the glass media for minimum 10 years.

COALLIFE



CARBON FILTERS

COALLIFE carbon filter is a purification device that uses activated carbon to remove impurities and contaminants from water or air through a process called adsorption. These highly porous filters trap particles, chemicals (like chlorine), volatile organic compounds (VOCs), Odors, and gases. They are commonly used in water purification.

In COALLIFE filters we are used advance activated carbon media for best performance and life long solution. The active carbon media is made from crushed coconut shell & IV900 value.

900 IV (Iodine Value) Activated Carbon is a high-performance, granular, or coconut shell-based filtration medium designed for superior water purification. It excels at removing organic pollutants, chlorine, pesticides, and unpleasant Odors, often utilized in 4x8 mesh sizes for water filtration systems.

AMR

METAL REMOVAL FILTERS



METALIFE Metal removal filters are specialized water treatment systems designed to eliminate toxic heavy metals (such as lead, arsenic, mercury, chromium) and nuisance metals (iron, manganese) from drinking or process water. These filters protect against long-term health risks and improve water taste, Odor, and appearance

In METALIFE filters we are used advance dmi65 media for best performance and life long solution. The DMI-65 protects and pre-treats all other water treatment systems from iron and manganese biofouling, effectively longer filter run times, and with the benefit of reduced oxidant demand and because DMI-65 Infusion technology requires no chemical regeneration.

Infusion Technology increases the catalytic surface and gives DMI-65 the highest oxidation rate and load capacity of any other catalytic media. Infusion Technology allows for very small attrition loss and displays no decaying effect or loss or catalysis for up to 5 – 10 years of continuous performance.

THANK YOU FOR YOUR TIME..!!

Feel Free to Contact us

AMR ENGINEERING WORKS

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